

VC_ARRAY

Overview

vc_array component is the model of the one-dimensional array of the multibit values. The size and the number of array elements is defined by corresponding configuration parameters.

Array built-in methods (tasks and functions) allow user to perform various operations on the array data, such as sorting, matching, reporting etc.

Syntax

```
vc_array #(bitw, size) <array_name> ();
```

Parameters		
bitw	Bit width of array data	
size	The size of the array	
Access Tasks		
Name	Arguments	Description
set	Index: integer Data: [bitw-1:0]	Defines the values of the array
swap	Index1: integer Index2: integer	Swaps two array elements, identified by their index numbers
shuffle	No Arguments	Randomly Shuffles all array values
sort	No Arguments	Sorts all elements in the array in the ascending order
Access Functions		
Name	Arguments	Description
get	Index: integer	Queries the values of the array using the array's index identifier.
Min_index	Dummy (0)	Returns the index of the smallest element in the array.
Min_data	Dummy (0)	Returns the value of the smallest element in the array.
Max_index	Dummy (0)	Returns the index of the largest element in the array.
Max_data	Dummy (0)	Returns the value of the largest element in the array.
match	Data: [bitw-1:0]	Compares given data with each one of the array elements. In the case of one or more matches, returns 1. Otherwise, returns 0
Match_index	Data: [bitw-1:0]	Returns the index of the matching data

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Verilog example

```
module test ();

integer number, i;

vc_array #(4,10) my_array ();

initial begin
    #10
    for (i=0; i<10; i=i+1)
        my_array.set(i,i);

    for (i=0; i<10; i=i+1)
        $display("Array data[%0d] = %0d", i, my_array.get(i));

    // Shuffling array
    // -----
    $display("\nShuffling an array...");
    my_array.shuffle;
    my_array.dumpd;

    // Minimum & Maximum data indexes
    // -----
    $display("\nMinimum & Maximum values...");
    $display("Min data index: %0d", my_array.min_index(0));
    $display("Min data value: %0d\n", my_array.min_data(0));
    $display("Max data index: %0d", my_array.max_index(0));
    $display("Max data value: %0d", my_array.max_data(0));
    my_array.dumpd;

    // Matching array values
    // -----
    $display("\nMatching an array...");
    if (my_array.match(6))
        $display("\nMatched index is %0d", my_array.match_index(6));

    // Sorting array values
    // -----
    $display("\nSorting an array...");
    my_array.sort;
    my_array.dumpd;

    #10 $finish;
end

endmodule
```